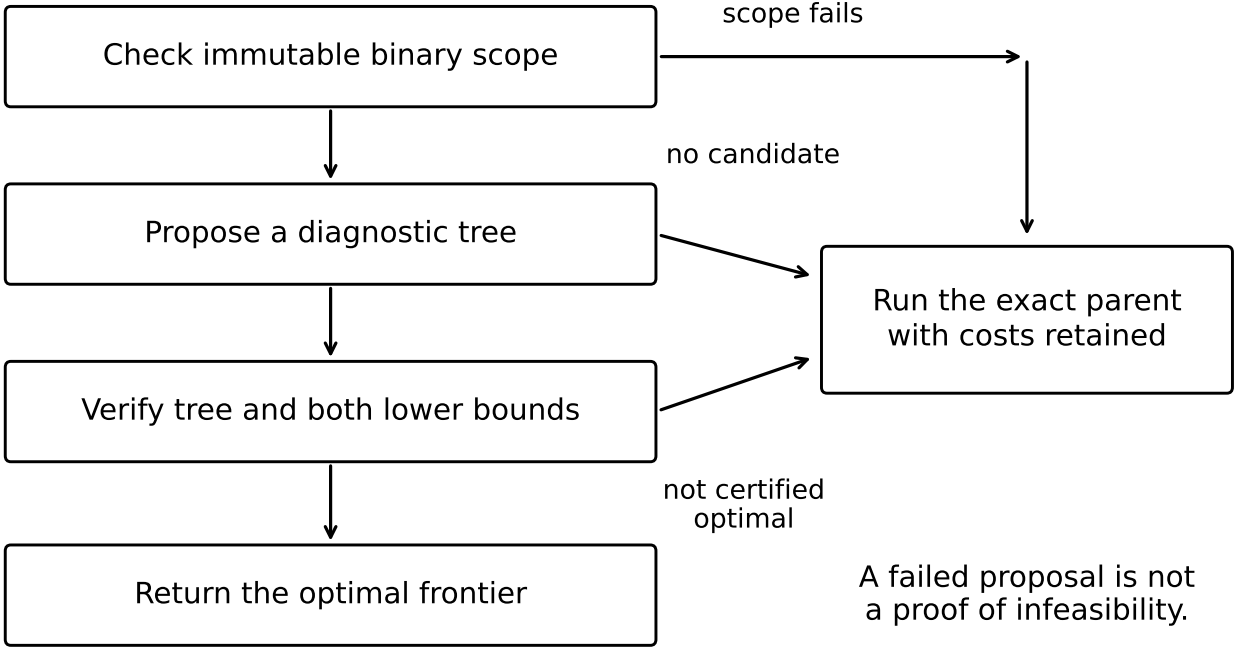
Check immutable binary scope
scope fails
Propose a diagnostic tree
no candidate
Verify tree and both lower bounds
Run the exact parent with costs retained
not certified optimal
Return the optimal frontier
A failed proposal is not a proof of infeasibility.